

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Duration: 60 Minutes

Total Marks: 25

Code of Conduct

I N.V. Sai Sahith (192425140) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N. Sahith

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Assessment Tasks

Short Answer Test

1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services..
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Short Answer Test Rubric

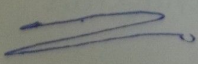
Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Evaluation from Standalone Computing to cloud Computing:-

the evaluation of computing began with Standalone Computers, where applications and data were stored and processed on a single machine. adopted client-server computing, allowing multiple users to access centralized Servers over a network.

the emergence of virtualization enabled multiple virtual machines to run on a single physical Server, increased resource utilization, High-Speed internet, broadband connectivity.

Key technological transitions:-

- standalone computing → client-server architecture.
 - client-server → distributed computing.
 - virtualization technology.
 - High-Speed internet and broadband.
 - SOA and web services.
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② Interpret the working of hypervisors in virtualization

A hypervisor is software that creates and manages (VMS) by sharing hardware resources among multiple OS.

Type-1 Hypervisor	Type-2 Hypervisor
Runs direct on hardware (bare metal)	Runs on the top of OS.
Higher performance and security	Easier to install a slighter slower
used in data centers and collocate.	used for testing personal use.
ex:- Xen, VMware ESXI.	ex:- Oracle, VMware work station.

③ Describe the role of APIs in cloud service intelligence

APIs allow different cloud applications and services to communicate with each other. They enable data sharing, automation, and integration b/w systems. APIs support interoperability by using standard protocols such as REST and SOAP, allowing services from different cloud providers to work together.

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Concept of resource pooling in Cloud Computing:-

Resource pooling means that cloud providers combine computing resources such as servers, storage, and networks into a shared pool. These resources are dynamically assigned to users based on demand.

This supports multi-tenancy, where multiple users share the same infrastructure securely. Benefits include better resource utilization, reduced cost, high availability, and efficiency scalability.

⑤ Concept of rapid elasticity in cloud Computing:-

Rapid elasticity is the ability of cloud systems to automate increase or decrease resources according to workload demand. During high traffic, more computing resources are added, and during low traffic, unused resources are removed. This ensures high performance, avoids resource wastage, reduces cost, and supports dynamic workloads.